

Mangrove crabs as ecosystem engineers; with emphasis on sediment processes

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Abstract

The benthic fauna in mangrove forests is usually dominated by burrowing sesarmid (Grapsidae) and fiddler crabs (Ocypodidae). They are herbivores that retain, bury, macerate and ingest litter and microalgal mats. Most species within these two groups actively dig and maintain burrows in the sediment as a refuge from predation and environmental extremes. Based on the current knowledge on the biology and ecology of these crabs, it seems obvious that their activities have considerable impact on ecosystem functioning. However, no convincing conceptual framework has yet been defined into which the role of these crabs can be identified and characterized. The attributes by which these abundant animals affect the microbial and biogeochemical functional diversity fit well into the concept of ecosystem engineering. The conceptualization of mangrove benthic communities within this framework is distinguished and documented by examples provided from the most recent literature on mangrove ecosystem functioning. It appears that the features and processes driving the engineering effects on distribution and activity of associated organisms operate differently for sesarmid and fiddler crabs. The most obvious and well-documented difference between engineering effects of the two types of crab seems to be associated with foraging. More attention must be devoted in the future to elucidate engineering aspects related to crab burrows in mangrove environments. Particularly comparative work on the burrow-dwelling life styles of the two types of crab is needed.

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1. Introduction

The mangrove ecosystem dominates along most tropical and subtropical coastlines (Por, 1984; Alongi, 2002). It is characterized as a forest growing in seawater at the interface between land and sea, particularly in sheltered areas with large tidal excursions. Mangrove forests are highly productive and support abundant planktonic and benthic communities, as well as high rates of microbial decomposition (Christensen, 1978; Alongi et al., 1999). The structuring feature of

mangrove forests is tree species adapted to growth in anoxic intertidal sediment saturated with seawater. Among their adaptations are special structures such as pneumatophores (e.g. *Avicennia* spp.) and prop roots (e.g. *Rhizophora* spp.) that supply the roots with oxygen from the air and provide physical support for growth in soft mud (Kitaya et al., 2002). Salt from the seawater is either excluded at the roots (e.g. *Rhizophora* spp.) or excreted by glands on the leaves (e.g. *Avicennia* spp.) (Passioura et al., 1992; Clarke and Allaway, 1993).

The benthic fauna associated with mangrove forests is typically dominated by various burrowing decapods, such as sesarmid crabs and fiddler crabs. They are

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herbivores that retain, bury, macerate and ingest litter and microalgal mats (Emmerson and McGwynne, 1992; Lee, 1997; Kristensen and Alongi, 2006). By doing so, they prevent loss of nutrients and promote decomposition processes. In addition, most sesarimid and fiddler crabs actively dig and maintain burrows in the sediment. These structures function as a refuge from predation and environmental extremes, and may in some cases provide for food storage (Giddins et al., 1986; Warren, 1990; Dittmann, 1996). The burrows affect sediment topography and biogeochemistry by modifying particle size distribution, drainage, redox conditions and organic matter as well as nutrient availability (Mouton and Felder, 1996; Botto and Iribarne, 2000).

Many studies have dealt with the role of benthic fauna in mangrove ecosystem functioning, and the ecological importance of crabs seems well recognized (Smith et al., 1991; Thongtham and Kristensen, 2003). However, no convincing conceptual framework has yet been defined to identify and characterize these features. Most ecosystem effects of crabs are caused by non-trophic activities and fit well into the concept of ecosystem engineering. They modify physical structures, transport conditions and substance chemistry, and by doing so change the availability of resources for the associated microbial, fauna and plant communities.

This contribution will identify and describe the engineering attributes by which crab species primarily belonging to the decapod subfamily Sesarminae (family Grapsidae) and fiddler crabs (genus *Uca*; family Ocypodidae) affect the availability of resources for other biological assemblages in mangrove systems. In the present context, these assemblages refer primarily to microbial communities that mediate biogeochemical reactions. The conceptualization of mangrove benthic communities within the framework of ecosystem engineering will be distinguished and documented from examples provided from the most recent literature on mangrove ecosystem functioning.

2. What are ecosystem engineers?

The concept of ecosystem engineering was introduced about a decade ago by Jones et al. (1994, 1997). They defined organisms as ecosystem engineers when they ‘directly or indirectly modulate the availability of resources (other than themselves) to other species, by causing physical state changes in biotic or abiotic materials. In so doing they modify, maintain and/or create habitats’. Direct transfer of resources by an organism to other species in the form of living or dead tissues is not engineering, but rather trophic interactions. Thus, a

critical characteristic of ecosystem engineers is that they must change the quality, quantity and distribution of resources utilized by other species, excluding the engineer itself as a resource. Two major types of ecosystem engineers can be defined dependent on the mechanisms by which they change the environment (Jones et al., 1994): autogenic engineers change the environment via their own physical structures (i.e. living and dead tissues), while allogenic engineers change the environment by transforming living or non-living materials from one physical state to another via mechanical or other actions.

Ecosystem engineering is not synonymous with the concept of ‘keystone species’. Keystone species control community structure (Paine, 1969; Krebs, 2001), and removal of keystone species causes massive changes in species composition and ecosystem functioning. The action of keystone species is usually regarded as trophic, but critical effects frequently involve engineering, for example via physical disturbance (Smith et al., 1991; Reise, 2002). It is as yet not fully clear whether the impact of many previously defined keystone species actually is due to engineering and not trophic effects.

Recently, Gutiérrez and Jones (2006) broadened the concept of ecosystem engineering to include organisms as agents of biogeochemical heterogeneity. These organisms can ‘affect biogeochemical reactions by changing the availability of resources for microbes (e.g. carbon and nutrients) or by changing abiotic conditions affecting microbial reaction rates (e.g. redox and temperature)’. Due to the nature of microorganisms, the engineering effects primarily occur at scales of millimetres, but may in many cases be extrapolated and averaged to ecosystem scale. In contrast to the ‘classic’ concept of ecosystem engineering (Jones et al., 1994, 1997), where the impact of engineering largely is related to species diversity changes within biotic communities, the biogeochemical approach also considers the functional diversity of microbial communities. While species diversity is an important determinant of ecological changes at the population and community level, functional diversity is essential for understanding changes with respect to element cycling at the ecosystem level. Gutiérrez and Jones (2006) argued that biogeochemical heterogeneity develops as a consequence of variations in either abiotic conditions or the availability of microbial resources. The heterogeneity and activity of microbial reactions in any patch are therefore controlled by processes that changes the physical environment (physical structure), transport conditions (physical transport), availability of materials (substances) and any combination of these (Fig. 1). The activity of allogenic ecosystem engineers can alter the characteristics of patches by forming structures (such as

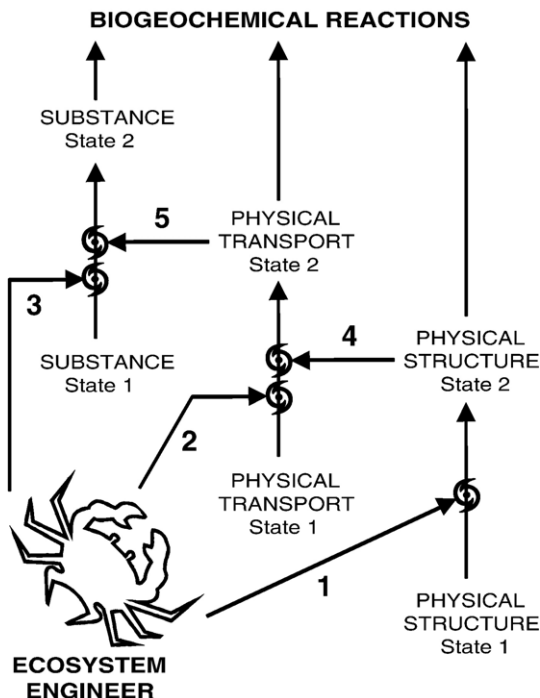


Fig. 1. Flow diagram showing the features by which ecosystem engineers affect biogeochemical activity and heterogeneity in sediment environments. The first feature includes modification of physical structures (for example burrows). The state change of the physical structure is driven by processes such as the burrowing and maintenance activity of the engineer (process 1). The second feature is related to physical transport (such as material, fluid and gas). Transport state changes means that the engineer redistributes materials by reworking within the sediment or actively pump fluid and gas in and out of the sediment via burrows (process 2). The physical structure itself can also passively mediate transport by processes such as collapsing burrows, passive fluid transport and diffusive gas transport (process 4). The third feature is associated with changes in the chemical properties of substances (for example, degradability and redox conditions). The engineer will typically handle or ingest potential food sources (substances) and by doing so modify and change the state of the substance (process 3). The modified substance (e.g. faeces) may become more or less available for other organisms, e.g. detritivores and decomposers. The physical transport will also potentially affect the state of substances by changing factors such as the redox conditions via removal of toxic metabolites and introduction of oxygen deep into sediments (process 5) (inspired by Gutiérrez and Jones, 2006).

burrows) that act as conduits or barriers for transport of organic matter, nutrients and redox active compounds (process 1, Fig. 1). The ecosystem engineer and the physical structures it forms can also affect the transport of substances to and from microbial communities by active transport of solids or fluids (processes 2 and 4, Fig. 1). Finally, the engineer can modify substances directly through e.g. handling (maceration), gut passage (enzymatic action) and burial by which the availability of these

substances to microbial communities is augmented or hampered (processes 3 and 5, Fig. 1).

Even before Jones et al. (1994) introduced the concept, many studies have shown excellent examples of ecosystem engineering related to sediment biogeochemistry. For example, the activities of large burrowing invertebrates in marine benthic environments have long been known to affect the physical and chemical structure of sediments, altering abiotic conditions and nutrient availability for biogeochemical processes (e.g. Aller, 1982; Berner and Westrich, 1985). Instead of allogenic ecosystem engineering, these features have been labelled by the broader and not particularly well-defined concept of bioturbation. In its strict sense, bioturbation refers to the physical displacement of particles and water caused by macrofaunal activities in sediments, but it does not directly refer to the associated impact on microbial drivers of biogeochemistry. The conceptual framework of ecosystem engineering, on the other hand, elegantly emphasizes the coupling between bioturbation and sediment biogeochemistry.

Although the ecological consequences of ecosystem engineering in many cases may be trivial or difficult to identify, it is of fundamental interest to identify the types of environments where ecosystem engineering play the most important role. Among such environments, the relatively unexplored mangrove ecosystem is an obvious candidate for future studies. The diversity of plants and animals in mangrove forests is generally low, but they all play important roles for ecosystem functioning. Trees are the key engineers that form the unique mangrove environment, but the crab fauna are responsible engineers that drive efficient element transfers by maintaining high functional diversity and biogeochemical heterogeneity in sediments.

3. Mangrove functional diversity and biogeochemical heterogeneity

Before focusing on the engineering impacts of crabs, the involved functional dynamics of microbial communities and biogeochemical reactions in mangrove sediments must be identified. Mangrove sediments are usually composed of fine-grained mud with up to about 80% of particles < 63 μm and an organic carbon content ranging from 1 to 6 mmol g^{-1} . The main source of organic matter in mangrove environments is surface-deposited litter from trees (leaves, propagules and twigs) and subsurface root growth (Alongi, 1998). The contribution of detritus from macroalgae, benthic microalgae, phytoplankton and epiphytes is generally low (i.e. < 14% of the total primary production, Alongi, 1998) in most mangrove forests

due to light limitation under the canopy. However, the $>10\times$ higher nitrogen content of algae (C:N ~ 7 –10) compared with mangrove litter (C:N ~ 100) implies that algae contribute with a large fraction of the organic nitrogen input to mangrove food webs. Stable isotopes have indicated that mangrove-derived detritus (tree litter) is an important food source for select groups of invertebrates, while other species primarily feed on algae (Newell et al., 1995; Bouillon et al., 2008-this issue).

Microbial decomposition is generally slow in mangrove sediments because most of the detritus is of poor nutritious value and contains high levels of retarding compounds. The detritus is rich in tannins, lignin and structural carbohydrates (cellulose), which inhibits microbial degradation (Lee et al., 1990; Benner et al., 1990; Marchand et al., 2005). While cellulose and lignin can readily be degraded in oxic environments, these compounds are only degraded slowly under anoxic conditions. Lignin, for example, has a half-life of >150 y in anoxic mangrove sediment (Dittmar and Lara, 2001). The formation of old and refractory material in mangrove sediments can be observed visually as lignified and humified (spongy) litter fragments. Accordingly, Dittmar and Lara (2001) estimated that the average age of organic carbon in the upper 1.5 m of the sediment in a Brazilian mangrove forest is between 400 and 770 y.

The decay of mangrove litter in sediments begins with leaching of soluble organic substances. Newly fallen litter loses 20–40% of the organic carbon by leaching when submerged in seawater (Camilleri and Ribi, 1986; Twilley et al., 1997). Much of the leached DOM is labile and rapidly consumed by the benthic microbial community (Kristensen and Pilgaard, 2001), while some is exported to the mangrove waterways. Further decomposition of the remaining particulate material occurs by extracellular enzymatic hydrolysis mediated by bacteria and mycelial decomposers followed by assimilation of the solubilized compounds. In contrast to other marine ecosystems where bacteria dominate degradation processes, marine mycelial decomposers belonging to eumycetes (Fungi) and oomycetes (Protoctista) are highly adapted for the capture of cellulose-rich vascular plant litter in mangrove ecosystems (Newell, 1996).

Rates of microbial carbon oxidation in mangrove sediments typically range from 20 to 60 mmol m⁻² d⁻¹ (Kristensen et al., 2000; Alongi et al., 2001). Aerobic respiration occurs near the sediment surface, around crab burrows and along oxic root surfaces. The consumption of O₂ at these interfaces is usually so rapid that O₂ rarely penetrates >1 mm into the sediment

(Kristensen et al., 1994). A wide variety of anaerobic microorganisms mediate most carbon oxidation below the oxic zone. Anaerobic sulfate reduction has long been considered the most important respiration processes in mangrove sediments (Alongi, 1998), with a share of up to 50% of the total carbon oxidation. Consequently, most mangrove sediments contain high levels of reduced inorganic sulfur primarily in the form of pyrite (Holmer et al., 1994). Recent evidence suggests, however, that the role of iron respiration for organic carbon mineralization may be comparable to or higher than that of sulfate reduction in iron-rich mangrove environments (Kristensen et al., 2000; Kristensen and Alongi, 2006). As sulfate reduction usually is hampered in the presence of more potent electron acceptors (e.g. O₂ and Fe³⁺; Canfield et al., 1993), this process becomes inferior to iron respiration when oxidizing roots and infaunal burrows increase the Fe³⁺ content deep in mangrove sediments (Nielsen et al., 2003). However, the role of each pathway varies considerably within and between mangrove environments, depending on several interrelated factors, including sediment composition, bioturbation, tidal inundation frequency, sediment composition as well as sulfate, iron and organic carbon availability (Table 1).

Sedimentary organic matter in mangrove environments is richer in nitrogen (molar C:N ratio of 20–30) than fresh litter (Kristensen et al., 1995). A rapid initial release of carbon and retention or enrichment of nitrogen must therefore occur during early degradation of mangrove litter (Twilley et al., 1997; Wafar et al., 1997). As microorganisms generally need substrates with an elemental C:N ratio <10 for maintenance and growth, mangrove litter is a highly insufficient substrate, even after burial into the sediment. Thus, a respiratory

Table 1
Percentage partitioning of electron acceptors for the terminal oxidation of carbon in mangrove environments

Location	e-acceptor				
	O ₂	NO ₃ ⁻	Mn ⁴⁺	Fe ³⁺	SO ₄ ²⁻
Thailand (+sesamid) ¹	3	1	~ 0	78	18
Thailand (+fiddler) ¹	56	~ 0	~ 0	33	12
Australia (+fiddler) ²	2	~ 0	~ 0	62	36
Tanzania (+mix) ³	14	~ 0	~ 0	77	9
Skidaway (+fiddler)	~ 0	~ 0	~ 0	95	5
Skidaway (-crabs) ⁴	~ 0	~ 0	~ 0	~ 0	100

The presence and absence of sesamid and fiddler crabs are indicated. The Skidaway saltmarsh data (*italics*) are included to emphasize the dramatic shift in microbial metabolism due to the presence of crabs. No similar type of study has been conducted in mangrove environments.

¹Kristensen et al. (2000); ²Kristensen and Alongi (2006); ³Kristensen et al. (unpubl. data) ⁴Kostka et al. (2002).

loss of ~75% of carbon from fresh litter is necessary to reach the observed sedimentary C:N ratio if other sources and sinks of carbon and nitrogen are excluded (Dittmar and Lara, 2001). Mangrove environments are therefore depleted in dissolved inorganic nitrogen (DIN) and the sediments generally act as sinks of DIN from the surroundings (Alongi, 1996; Rivera–Monroy and Twilley, 1996). Accordingly, the cycling of inorganic nitrogen must be rapid and efficient (i.e. tight) to support microbial growth in mangrove sediments.

4. Engineering effects of sesarmid and fiddler crabs on mangrove biogeochemistry

4.1. Burrow construction and maintenance

Most sesarmid and fiddler crabs construct and maintain burrow structures in mangrove sediment with significant engineering effect (processes 1, 2 and 4, Fig. 1) on the distribution and activity of associated organisms. The burrows are used by the crabs as a retreat or refuge when environmental conditions at the sediment surface are intolerable, for example during high tide or when conditions are too dry and hot (De la Iglesia et al., 1994; Botto and Iribarne, 2000; Thongtham and Kristensen, 2003). The two groups of crab generally occupy different habitats within the mangrove environment. Sesarmids are most abundant under tree canopies where their food source, leaf litter, is abundant, while the network of pneumatophores and cable roots provide protection against predators and extreme temperatures (Thongtham and Kristensen, 2003). Fiddler crabs, on the other hand, prefer open areas, particularly near creek banks, where strong sunlight stimulates growth of microphytobenthos, which is a primary food source for these crabs (Nobbs,

2003). Furthermore, they avoid any harassment and predation by sesarmids in these areas. Both types of crab clearly modify their habitat and create much of the visible topographic structures, e.g. mounds and depressions that are typical for mangrove sediments (Warren and Underwood, 1986).

The morphology of fiddler crab burrows is quite simple and similar among species, and it usually consists of a more or less permanent vertical shaft extending 10 to 40 cm into the sediment (Fig. 2A). Often, the burrows are bent into a J- or L-shape ending with a chamber (Koretsky et al., 2002; Lim, 2006). The burrows are continuously constructed, maintained and abandoned. The amount of sediment excavated during burrow construction and maintenance is considerable. Thus, McCraith et al. (2003) estimated that a population of *Uca pugnax* (40–300 burrows m⁻²) in a North American saltmarsh excavated 120–160 cm³ of sediment m⁻² d⁻¹ and continually mixed the upper 8–15 cm of the sediment. However, the excavated sediment must be counteracted by infilling of abandoned burrows and collapse of burrow walls. As a result, fiddler crabs alter the quality of organic matter on the sediment surface by replacing surface derived reactive material (e.g. fresh microphytobenthos) with much less reactive and partly degraded material from depth in the sediment (Gutiérrez et al., 2006).

Burrows of sesarmid crabs vary considerable in morphology and some species do not construct burrows, but utilize crevices or abandoned burrows of other species (Gillikin and Kamanu, 2005). Burrows of *Neoepisesarma versicolor* in Thailand, for example, range from very simple, straight burrows with few branches to complex, labyrinthine structures with up to 5 openings (Fig. 2B) (Thongtham and Kristensen, 2003).

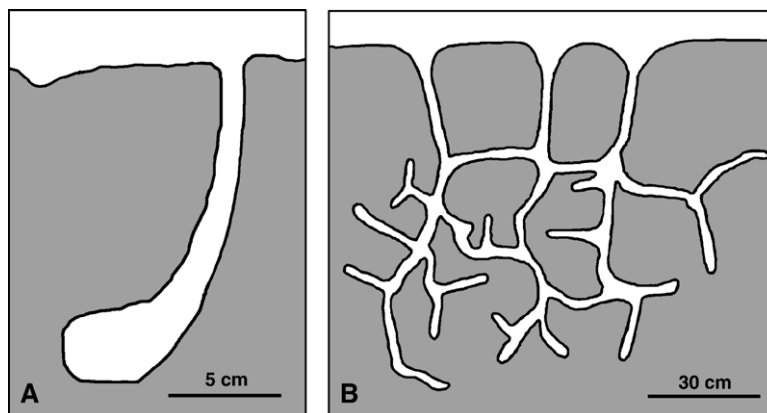


Fig. 2. Schematic drawings showing (A) the typical shape and dimensions of fiddler crab burrows, and (B) sesarmid crab burrows (modified from Kristensen and Kostka, 2005).

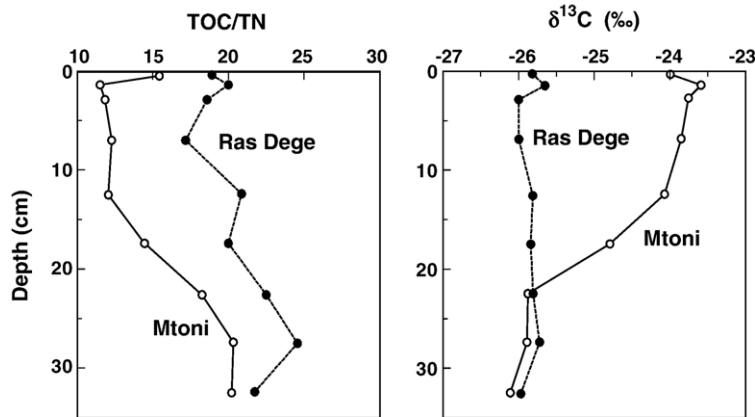


Fig. 3. Vertical profiles of TOC/TN ratio and $\delta^{13}\text{C}$ in the *Ulva lactuca* impacted Mtoni mangrove forest and the pristine Ras Dege mangrove forest. The data were obtained during the dry season in September (Bouillon and Kristensen, unpubl. data).

The surface opening of *N. versicolor* burrows is always associated with a more or less vertically oriented shaft. The length of the shaft varies from 5 to 25 cm with small burrows belonging to an associated fauna of small crustaceans and juvenile sesarmids protruding as side branches. The deeper part of *N. versicolor* burrows usually comprises smooth walled horizontal tunnels with several dead-ends and small chambers in an unpredictable pattern. The depth of burrows varies from ~55 to 110 cm. Crab burrow morphology is affected by factors such as sediment characteristics and age of burrows as well as type and abundance of associated fauna. Sesarmid crab burrows are long-lived and particularly the deeper parts are expected to last for a very long time. The complex structure of these deeper parts may actually be the result of continued construction by several generations of occupants. Because of frequent collapse of the near-surface parts of burrows, the crabs must continually maintain, rearrange and often reposition entrances (Micheli et al., 1991). Although the abundance of sesarmid crabs is low ($1\text{--}5 \text{ ind m}^{-2}$) compared with fiddler crabs, these larger crabs excavate an equivalent amount of sediment ($80\text{--}210 \text{ cm}^3 \text{ m}^{-2} \text{ d}^{-1}$) and probably affect the organic matter composition at the sediment surface to at least the same extent.

In contrast to fiddler crabs, many sesarmid crabs tend to bury leaf litter and detritus during burrow construction and maintenance. The litter is frequently pulled directly into the burrows (Robertson, 1986; Ashton, 2002; Skov et al., 2002). The extent by which sesarmid crabs translocate deposited litter is clearly evident from the results of a recent study near Dar es Salaam, Tanzania (Bouillon and Kristensen, unpublished; see also Bouillon et al., 2008-this issue). The eutrophic Mtoni mangrove area receives masses of drifting macroalgae (e.g. *Ulva* sp.) via tidal

action, particularly during spring tides in the dry season. Algae deposited at the sediment surface are handled, ingested and ultimately buried by sesarmid crabs. No such algal deposition occurs in the nearby oligotrophic Ras Dege mangrove area, where crabs primarily handle mangrove litter. The chemical composition of macroalgal detritus differs from mangrove litter with respect to TOC/TN ratios and $\delta^{13}\text{C}$ signatures (macroalgae: TOC/TN ~7–15, $\delta^{13}\text{C}$ ~–15 ‰; mangrove litter: TOC/TN ~50–100, $\delta^{13}\text{C}$ ~–30 ‰). The intensive subduction of macroalgal detritus by sesarmid crabs at Mtoni is clearly evident as lower TOC/TN ratio and enriched $\delta^{13}\text{C}$ signals of the bulk organic matter in the sediment down to 20–30 cm compared with Ras Dege, where leaf litter is a more important organic source (Fig. 3).

Except for occasional litter plastering by sesarmid crabs, neither of the two groups of crab directly stimulate

Table 2

Abundance and basic dimensions of various sesarmid and fiddler crab burrows

Species	Abundance	Depth	Surface area	Reference
	ind. m^{-2}	cm	m^2/m^2	
<i>Uca pugilator</i>	52 ± 10	15 ± 10	0.46	1
<i>Uca pugnax</i>	13 ± 4	20 ± 3	0.15	1
<i>Uca vocans</i>	12	10 ± 3	0.08	2
<i>Uca annulipes</i>	12	10 ± 3	0.07	2
<i>Neopisesarma versicolor</i>	0.2	84 ± 7	0.33	3
<i>Sesarma messa/</i> <i>Alpheus cf. macklay</i>	1.7	120	6.46	4

The inner burrow wall surface area (m^2) is given per m^2 sediment surface for the entire population.

1: Koretsky et al., 2002; 2: Lim, 2006; 3: Thongtham and Kristensen, 2003; 4: Stieglitz et al., 2000.

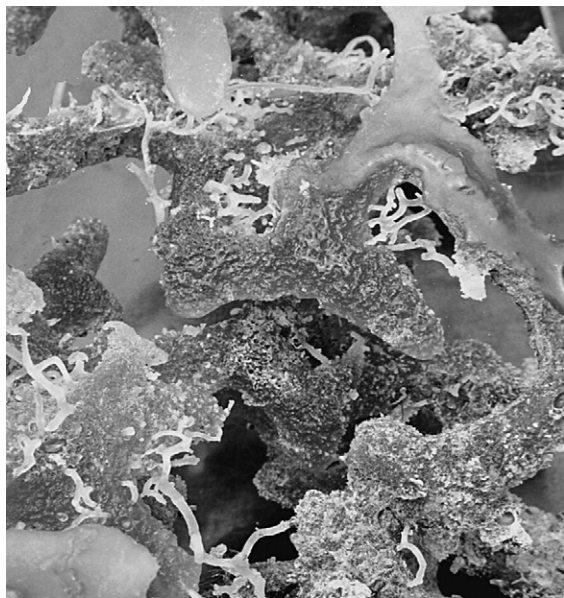


Fig. 4. Close-up photo of burrow resin cast of the sesarmid crab *Neopisesarma versicolor* from the Bangrong mangrove forest, Phuket, Thailand. The numerous small burrows of the associated fauna are clearly visible as fine protrusions along the main burrow shaft (Thongtham and Kristensen, unpubl. data).

microbial activity along burrow walls by secreting reactive mucus (Kristensen and Kostka, 2005). Their burrowing activities nevertheless increase the potential for biological and chemical exchange simply by increasing the interface between sediment and water/air. According to the data available (Table 2), each fiddler crab burrow increases the interface area by $\sim 1\%$, while the increase per sesarmid crab burrow is 150–380%. If the up to two orders of magnitude differences in abundance between the two groups are considered (fiddler crabs: $10\text{--}500\text{ m}^{-2}$; sesarmid crabs: $0.2\text{--}10\text{ m}^{-2}$), the increase in interface area under natural conditions is considerable and comparable for both (Hartnoll et al., 2002; Skov et al., 2002; Thongtham and Kristensen, 2003).

The gas and solute exchange across the walls of the numerous burrows can only be rectified if there is access to transport via air or water phases. In contrast to the engineering effect of many subtidal polychaetes, as pointed out by Gutiérrez and Jones (2006), no mangrove crabs actively ventilate their burrows. Only their sporadic crawling movements into and out of as well as inside burrows may mediate some transport. In fact, many fiddler crabs even prevent transport by plugging their burrow entrances during tidal inundation because it allows them access to trapped air suitable for respiratory needs and prevents burrow collapse (De la Iglesia et al.,

1994). Both sesarmid and fiddler crab burrows are nevertheless dependent on passive irrigation via tidal action, but the mechanism by which water and air are transported and thus the functionality of burrows depends strongly on the sediment characteristics and burrow structure. Burrows in permeable sandy sediments are usually flushed with water and air through tidal-driven percolation within the sand, while deep burrows in cohesive sediments are flooded from above during high tide and drain through deep cracks or by pressure differences between multiple openings (Thongtham and Kristensen, 2003). Stieglitz et al. (2000) found that the total flow through large burrows of *Sesarma messa* in northern Australia is up to $100\text{ L m}^{-2}\text{ tide}^{-1}$, which is equivalent to 30% of the tidal water entering that particular mangrove area. Along with the water drainage during low tide, oxygen is introduced deep (i.e. $>1\text{ m}$) into the sediment via air intrusion causing cascading redox effects on metal and sulfur cycling.

The engineering impacts of burrows as physical structures and associated physical transport on small fauna and microorganisms in the sediment are numerous and widespread. It may, however, at times be difficult to discriminate the effects of the structures themselves from those mediated by transport. The association of small burrowing fauna (juvenile crabs, callianassids, polychaetes and gastropods) within burrows of adult sesarmid crabs (Thongtham and Kristensen, 2003) is a clear example of ecosystem engineering caused by physical structures (i.e. burrows) (Fig. 4). The adult crabs modify the environment by burrow construction (process 1, Fig. 1) favouring the occupation of new ecological niches where small fauna have access to constant environmental conditions, ample food supplies and protection from predators. No such association has been recorded for the relatively short-lived fiddler crab burrows, whereas it frequently occurs in e.g. thalassinid burrows (Kinoshita, 2002; De Gibert et al., 2006). It is likely that meiofaunal abundance and community structure also is altered along burrow walls of both sesarmid and fiddler crab burrows as reported for polychaete

Table 3

Fluxes ($\text{mmol m}^{-2}\text{ d}^{-1}$) across the sediment-water/air interface of CO_2 , NH_4^+ and NO_3^- in the Mtoni and Ras Dege mangrove forests, Tanzania (Kristensen, unpubl. data)

Compound	Mtoni	Ras Dege
CO_2	106.3 ± 18.3	78.1 ± 7.2
NH_4^+	2.03 ± 0.20	-0.91 ± 0.29
NO_3^-	1.12 ± 0.59	-0.29 ± 0.02

Results are given as mean $\pm$ SE ($n=3$). Negative values indicate uptake by the sediment.

burrows in temperate areas (Reise, 1981), but no reports are yet available to confirm this proposition for mangrove environments.

The most obvious engineering impact caused by crab-mediated physical transport (process 2, Fig. 1) is the downward translocation of fresh litter detritus and the associated influence on heterotrophic microbial communities within the sediment. It has been argued that the relocation of microphytobenthic and macrophyte detritus by the excavation activities of crabs or litter directly pulled into burrows enhances the heterogeneity and thereby the efficiency of microbial decomposition in subsurface mangrove sediments (Emmerson and McGwynne, 1992; Micheli, 1993; Botto and Iribarne, 2000). The study from Tanzania mentioned above, where a massive input of *Ulva* sp. is buried by sesarmid crabs in the Mtoni mangrove forests, clearly shows a subsequent impact of detritus burial on bulk sediment carbon and nitrogen cycling (Table 3). The labile and nitrogen-rich *Ulva* detritus in Mtoni mediates rapid carbon and nitrogen mineralization within the sediment that result in large efflux of CO_2 and dissolved inorganic nitrogen (NH_4^+ and NO_3^-). Conversely, at Ras Dege where the sediment is supplied with more refractory and nitrogen-poor leaf litter, the lower efflux of CO_2 suggests an overall modest microbial activity. Furthermore, the uptake of dissolved inorganic nitrogen by this sediment complies with the general notion of nitrogen limitation in mangrove sediments when leaf litter is the primary source of organic matter (Alongi, 1996; Kristensen, unpubl. data).

Another important engineering effect of sesarmid and fiddler crab burrows is related to the biogeochemistry of redox-sensitive and largely microbially-driven processes involving iron and sulfur. A number of studies have in recent years suggested that macrofauna substantially affect the geochemical cycles of S and Fe in marine sediments (e.g. Kostka et al., 2002; Gribsholt et al., 2003; Nielsen et al., 2003). In the case of mangrove crabs, formation of burrow structures (process 1, Fig. 1) and the largely passive transport of water and air channelled by these (process 4, Fig. 1) are involved mutually. The driving factor is the intrusion of oxygen that enhances aerobic decomposition (Kristensen and Holmer, 2001), increases the volume of oxidized Fe^{3+} -containing sediment, and neutralizes toxic metabolites from anaerobic processes (e.g. sulfide). Crab burrows may therefore affect the partitioning between electron acceptors (e.g. SO_4^{2-} versus Fe^{3+}) used by anaerobic bacteria in the terminal oxidation of organic carbon (Table 1). The key to maintaining a delicate redox balance is availability of a sufficiently large reactive iron

pool within the sediment. Radial profiling of crab burrows in iron-rich mangrove sediments has confirmed an increased supply of reactive Fe^{3+} along the walls, which causes Fe^{3+} reduction to out-compete sulfate reduction (Fig. 5). Thus, crab burrows (and plant roots) maintain sufficiently high redox conditions to prevent excessive rates of sulfate reduction and associated accumulation of toxic sulfide in mangrove sediment. The sulfide produced under these conditions is largely precipitated

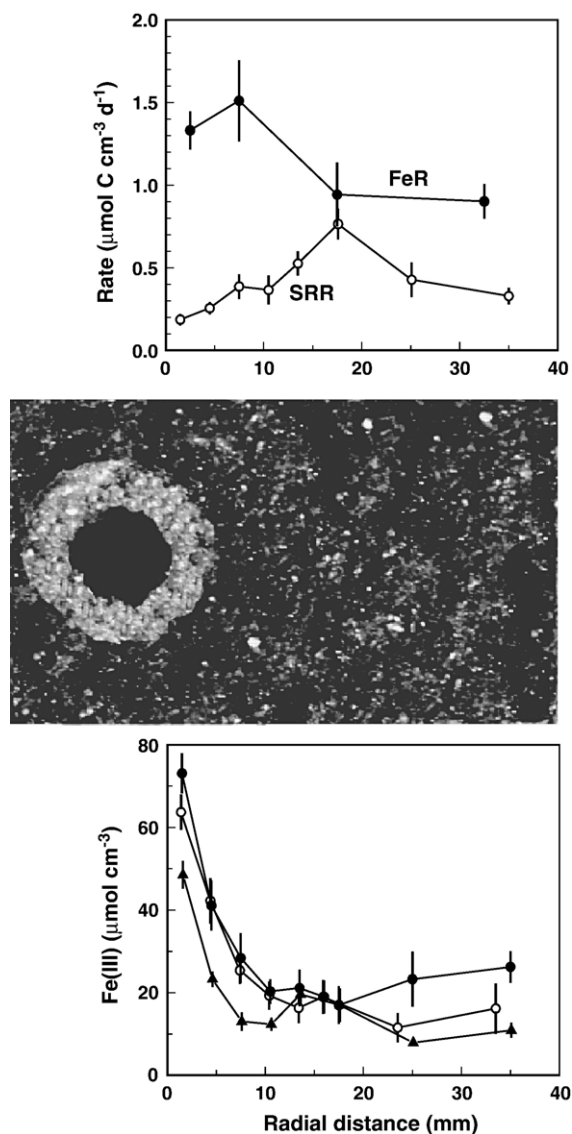


Fig. 5. Radial profiles around a fiddler crab burrow of: Upper panel, dissimilatory iron reduction (FeR) and sulfate reduction (SRR), and lower panel, oxidized iron (Fe^{3+}). The middle photograph shows the radial colour changes around the burrow. The light oxidized zone extends 5–10 mm away from the burrow lumen and into the surrounding sediment. Further away, black mackinawite (FeS) dominates the coloration of the sediment (modified from Gribsholt et al., 2003).

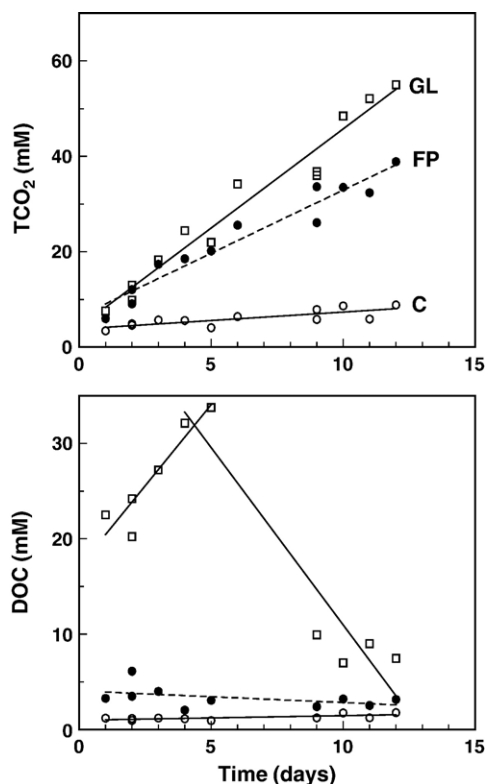


Fig. 6. Temporal pattern of TCO₂ and DOC concentrations in anoxic sediment incubations amended with green *Rhizophora apiculata* leaves (GL), faecal pellets from the sesarmid crab *Neoepisesarma versicolor* after feeding on green *Rhizophora apiculata* leaves (FP). The control sediment (C) was kept unamended (modified from Kristensen and Pilgaard, 2001).

as relatively inert pyrite (Holmer et al., 1999), which is harmless to plants and animals.

4.2. Sesarmid crabs and the fate of ingested plant litter

Burrow construction and maintenance is not the only engineering effect of mangrove crabs. Also the handling and ingestion of food substances (processes 3 and 5, Fig. 1) has substantial effects on other organisms in the environment. Many sesarmid crab species are remarkable in their ability to consume mangrove litter (Dye and Lasiak, 1987; Robertson et al., 1992). Studies from a variety of mangrove environments have shown that litter consumption and burial by sesarmid crabs may remove 30–90% of the annual litterfall (e.g. Robertson, 1986; Micheli, 1993; Slim et al., 1997). It appears that the senescence and degradation stage of leaves is important to their palatability to crabs. Thus, Thongtham and Kristensen (2005) showed that *Neoepisesarma versicolor* consumes fresh green and senescent yellow *Rhi-*

zophora apiculata leaves at a considerably lower rate than partly degraded brown leaves. This observation is consistent with those of Giddins et al. (1986) for *Neosarmatium smithii* and Micheli (1993) for *Sesarma messa* and *N. smithii*. The higher palatability of brown leaves compared with green and, in particular, yellow leaves probably results from improved nutritional value and removal of inhibitory compounds by the aging process (Poovachiranon and Tantichodok, 1991).

In general, sesarmid crabs consume about half of the handled litter immediately, while the remainder is pulled into burrows to promote microbial colonization and leaching of tannins. The litter stored in burrows may later be consumed by the crabs or is left for further microbial degradation. After shredding and maceration followed by passage through the crabs intestinal system, much of the litter is returned to the sediment environment in the form of partly digested faeces. The assimilation efficiency of carbon by sesarmid crabs feeding on a leaf diet ranges from 40 to 70% (Lee, 1997; Thongtham and Kristensen, 2005), which means that about half of the ingested litter subsequently becomes available for decomposer or detritus food webs. It is noteworthy that the C:N ratio of faeces is lower than in the eaten leaf litter (Lee, 1997; Thongtham and Kristensen, 2005), probably due to bacterial growth and nitrogen retaining within the intestinal system.

Kristensen and Pilgaard (2001) provided evidence for the engineering effect of deposited faecal pellets from *N. versicolor* fed green *R. apiculata* leaves on the heterogeneity of anaerobic microbial activity in sediment. They compared the fate of carbon in chopped green leaves (less than 3 by 3 mm) with that of freshly collected faecal material after the detritus was buried in anoxic mangrove sediment. Anaerobic microbial decay determined as TCO₂ production was 11.7× higher for sediment amended with green leaf detritus than without any organic matter addition, while the stimulation in sediment amended with faecal material only was 7.5 times (Fig. 6).

Table 4

Rates of anaerobic carbon transformations (nmol cm⁻³ d⁻¹) in mangrove sediment after homogeneous mixing with 0.5% (dw) green leaf detritus (GL) of *Rhizophora apiculata* or 0.5% (dw) faecal pellets (FP) of *Neoepisesarma versicolor* after feeding on green *Rhizophora apiculata* leaves

	TCO ₂ prod	DOC prod	Net C prod
C	277±66	37±13	314
GL	3243±215	-2899±464	344
FP	2068±169	-95±67	1973

The control sediment (C) was handled similarly, except for the addition of organic matter. Values are given as mean±SE as obtained from Fig. 6 of Kristensen and Pilgaard (2001).

At first glance, it appears that the uneaten leaf detritus sustained highest microbial activity, but a closer look at the data reveals that excessive leaching of labile DOC from the fresh leaf detritus material (up to 30 mM in the porewater) supported most of the excess microbial activity in this treatment. Thus, after correction for the differential contribution of DOC to the microbial carbon oxidation, it turned out that the solid faecal material supported 55× faster microbial decay than the solid leaf material (Table 4). Lee (1997) similarly showed that faecal material from *S. messa* feeding on *R. stylosa* is a profitable food source for the detritus-feeding amphipod *Parhyalella* sp. Individuals of *Parhyalella* sp. fed with detritus containing crab faeces attained significantly higher moulting frequency and lower mortality than those fed a diet of leaf detritus alone.

4.3. Fiddler crab forcing of the microbial distribution and activity in sediment

The engineering effects of fiddler crab foraging are generally confined to the upper few millimetres of the sediment (Dye and Lasiak, 1986). They feed on fine particles by picking sediment from the surface using the minor chelae and placing it in the mouth cavity. Here the edible particles including microalgal cells, nematodes, and bacteria are sorted and ingested. The non-edible particles are shaped into small irregular balls that are deposited at the sediment surface at regular intervals by the minor chelae (France, 1998). There has been some controversy in the literature about the food selectivity of fiddler crabs. Some authors argue that *Uca* species primarily select bacteria (Dye and Lasiak, 1986), while

others claim that the major food source is microphytobenthos (France, 1998; Reinsel, 2004). There is obviously different selectivity among species and the outcome strongly depends on environmental conditions (Meziane et al., 2002). For example, Kristensen and Alongi (2006) found that grazing by the fiddler crab, *Uca vocans*, maintained near-surface microalgal abundance (Chl-a) 64% lower and bacterial abundance 23% lower than in defaunated control sediment (Table 5). While most species of fiddler crabs clearly ingest both benthic microalgae and bacteria (France, 1998; Bouillon et al., 2002), it is less clear how they handle meiobenthos and small macrobenthos (Reinsel, 2004). The commonly observed reduction of meiofaunal density and diversity by fiddler crabs (Schrijvers et al., 1995; Olafsson and Ndaro, 1997) can be a direct effect through predation or an indirect effect through disturbance or competition for food (Reinsel, 2004; Weis and Weis, 2004).

While effects of fiddler crab feeding are largely confined to the sediment surface, their activities may affect redox sensitive elements (e.g. Fe) as well as engineering effects of microbial distribution and activity down to at least 2 cm depth. The 35–230% higher content of Fe³⁺ and 50–90% lower content of Fe²⁺ found by Kristensen and Alongi (2006) in the upper 2 cm of the sediment when *Uca vocans* is present (Table 5) must be caused by continuous mixing and oxidation of surface sediment by the intense activities of the crabs. Mixing occurs not only during feeding but also when the dactyls of the legs sink into the sediment during crawling. Deposition at the surface of feeding balls and subsurface sediment during burrow

Table 5

Vertical distribution of organic carbon content (POC), microphytobenthos (Chl-a), bacteria, reactive Fe³⁺ (Fe(III)), reactive Fe²⁺ (Fe(II)), total carbon oxidation (C-ox), dissimilatory iron reduction (FeR) and dissimilatory sulfate reduction (SRR) in *Avicennia marina* vegetated mangrove sediment in the presence (+C) and absence (–C) of the fiddler crab, *Uca vocans*

	0–1 cm		1–2 cm		6–8 cm	
	+C	–C	+C	–C	+C	–C
POC (mmol g ^{–1})	1.51±0.05	1.83±0.12	1.52±0.03	1.68±0.03	1.40±0.27	1.58±0.01
Chl-a (µg g ^{–1})	2.9±0.5	8.0±0.6	2.4±0.2	3.7±0.8	2.3±0.4	2.0±0.3
Bacteria (10 ⁹ cm ^{–3})	6.1±0.5	7.9±0.7	5.5±0.3	6.3±0.9	4.2±0.9	4.2±0.2
Fe(III) (µmol cm ^{–3})	115±4	85±5	75±16	23±3	12±5	9±1
Fe(II) (µmol cm ^{–3})	2±0	20±1	45±15	94±5	124±9	132±8
	0–2 cm				6–8 cm	
	+C	–C			+C	–C
C-ox (µmol cm ^{–3} d ^{–1})	0.29±0.00	1.43±0.00			0.34±0.13	0.43±0.12
FeR (µmol cm ^{–3} d ^{–1})	0.43±0.16	6.56±1.25			0.41±0.63	0.14±0.63
SRR (nmol cm ^{–3} d ^{–1})	8±2	32±7			111±32	82±30

Values are given as mean±SE (n=3) (data from Kristensen and Alongi, 2006).

construction and maintenance, combined with the smoothening and oxidation effect of crawling crabs, augments the effective mixing depth. The low rates of carbon oxidation in near-surface sediment inhabited by *U. vocans* (Kristensen and Alongi, 2006; Table 5) is caused by lack of reactive organic carbon and not availability of electron acceptors (O_2 , Fe^{3+} or SO_4^{2-}). This is substantiated by the 15x higher dissimilatory iron reduction (FeR) in the surface of ungrazed mesocosms that are poorer in Fe^{3+} but much richer in labile microalgal organic carbon. The parallel, but much less dramatic, pattern with 4x higher sulfate reduction (SRR) near the sediment surface in ungrazed mesocosms compared with grazed mesocosms is probably also caused by the higher organic matter availability.

5. Conclusions: Are mangrove crabs really ecosystem engineers?

The answer to this question must be a clear and unambiguous, yes! While this is said, it must be emphasized that the features and processes driving engineering effects on the distribution and activity of associated organisms in general are operating differently for sesarmid and fiddler crabs (Table 6). This is not surprising considering the divergent biology of the two types of crab with respect to factors such as preferred habitats, complexity and size of burrow morphologies, and food preferences.

It is well documented that burial of plant detritus by the excavation activities of sesarmid crabs or litter directly pulled into their burrows enhance the heterogeneity and thereby the efficiency of microbial decomposition in subsurface mangrove sediments. No such

information is yet available for fiddler crabs. Also the passive irrigation of both sesarmid and fiddler crabs with water and air that introduces oxygen deep into the sediment enhances aerobic decomposition as well as modifies the partitioning between electron acceptors in the terminal oxidation of organic carbon. The intensity and depth distribution by which these engineering effects operate may vary considerably between the two types of crab. Burrow construction may also favour the occupation of new ecological niches, where small fauna (juvenile macrofauna and meiofaunal) have access to constant environmental conditions, ample food supplies and protection from predators. This engineering effect is well known for other burrowing marine invertebrates, but has only been proved sporadically for sesarmid crabs and not at all for fiddler crabs. More attention must certainly be devoted in the future to fully elucidate the engineering aspects related to crab burrows in mangrove environments. Particularly comparative work on the burrow-dwelling life styles of sesarmid and fiddler crabs is needed.

The most obvious and well-documented difference between engineering effects of the two crab groups seems to be associated with foraging. The litter handling and ingestion by sesarmid crabs form finely fragmented organic material that is an ideal substrate for and stimulate both microbial colonization and macrofaunal detritivores. Conversely, the surface-feeding activities of fiddler crabs efficiently remove reactive organic carbon (e.g. microphytobenthos and bacteria) and thereby diminish the biological activity in surface sediment by ingesting and competitively starving decomposers and other organisms involved in the detritus food web.

Table 6

Overview of ecosystem engineering attributes of sesarmid and fiddler crabs and their effect on the associated biological assemblages

Feature and state change	Biological/biogeochemical effect	Crab responsible
Burrow construction and maintenance	Creates ecological niches for small fauna	Sesarmid (and fiddler)
Redistribution of organic matter and burial of litter	Enhanced heterotrophic microbial activity in sediment	Sesarmid (and fiddler)
	Increased microbial heterogeneity	
	Nutrient conservation	
Exchange between sediment and water/air by:	Changed subsurface sediment redox conditions	Sesarmid and fiddler
Passive irrigation	Enhanced subsurface microbial iron reduction	
Diffusive gas exchange	Lowered microbial sulfate reduction	
Litter feeding and egestion	Enhanced heterotrophic microbial activity in sediment	Sesarmid
	Nutrient conservation	
	Improved nutrition for detritivores	
Surface feeding and mixing	Changed near-surface redox conditions	Fiddler
	Increased availability of oxidized iron	
	Lowered microbial biomass	
	Lowered near-surface heterotrophic microbial activity	
	Enhanced subsurface microbial iron reduction	

Parentheses indicate that effects are likely for fiddler crabs, but that no evidence currently exists.

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